

## Functions of Several Variables:

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- function which several input variables and one or more output variables.

Ex: following are real valued functions of two variables  $x, y$

①  $f(x, y) = x^2 + y^2$  over  $\mathbb{R}^2$

②  $f(x, y) = \frac{xy}{x^2 + y^2}$  defined over  $\mathbb{R}^2 / \{(0, 0)\}$

### Some applications

- ① Temperature distribution in a medium is a real valued function with more than 2 variables. Temperature function at time  $t$  and at point  $(x, y)$  has three variables.

Ex the temp. distribution in a plate (unit square) with 0 temp. at the edges and initial temp (at time  $t=0$ )

$T_0(x, y) = \sin \pi x \sin \pi y$  is  $T(t, x, y) = e^{-\pi^2 k t} \sin \pi x \sin \pi y$

- ② Sound waves and water wave problems in physics

$$u(x, t) = A \sin(kx - \omega t).$$

represents the travelling wave of the initial wave front  $\sin kx$ .

### Some useful Definitions:

- $\mathbb{R}^2 = \{(x, y) : x, y \in \mathbb{R}\}$

- Open ball of radius  $r$  with center  $(x_0, y_0)$  is

$$B_r(x_0, y_0) = \{(x, y) : \sqrt{(x - x_0)^2 + (y - y_0)^2} < r\}$$

- Open ball of radius  $\delta$  and center  $(0,0)$  is

$$B_\delta(0,0) = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 < \delta^2\} \text{ is open set}$$

- $\overline{B_\delta(0,0)} = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 \leq \delta^2\}$  is close set.

Limit of func<sup>n</sup> of Several Variables:

(Simultaneously/Double limit)

Let  $\Omega$  be an open set in  $\mathbb{R}^2$ ,  $(a,b) \in \Omega$  and let  $f$  be a real valued function defined on  $\Omega$  except possibly at  $(a,b)$  then the limit

$$\lim_{(x,y) \rightarrow (a,b)} f(x,y) = L$$

if for any  $\epsilon > 0 \exists \delta > 0$  s.t.

$$\sqrt{(x-a)^2 + (y-b)^2} < \delta \Rightarrow |f(x,y) - L| < \epsilon$$

Note:

If limit exists <sup>then it is</sup> ~~and~~ unique.

that is the limit is independent of choice of path chosen  $(x,y) \rightarrow (a,b)$

Some Examples of func<sup>n</sup> where limit does not exist

Ex 1 Show ~~limit~~  $f(x,y)$  does not exist for  $f(x,y) = \frac{x^2 - y^2}{x^2 + y^2}$

Soln We will show limits along different path ~~does not~~ are not equal thus not exist.

Along x-axis i.e. along  $y=0$

$$f(x,y) = f(x,0) = \frac{x^2}{x^2} \Rightarrow \lim_{x \rightarrow 0} f(x,0) = \frac{x^2}{x^2} = 1$$

Along y axis i.e.  $x=0$

$$f(x, y) = f(0, y) = -\frac{y^2}{y^2}$$

$$\Rightarrow \lim_{y \rightarrow 0} f(0, y) = -1$$

Limit DNE.

Another approach for the same example:

Along any arbitrary line  $y=mx$

$$f(x, y) = f(x, mx) = \frac{x^2 - m^2 x^2}{x^2 + m^2 x^2} = \frac{1 - m^2}{1 + m^2}$$

$$\lim_{x \rightarrow 0} f(x, mx) = \frac{1 - m^2}{1 + m^2}$$

for different value of  $m$ , we have different limits.  
so limit DNE.

Ex2 Does the limit of  $f(x, y) = \frac{xy^2}{x^2 + y^4}$  as  $(x, y) \rightarrow (0, 0)$  exist and if yes then what is the value?

Ans  $\lim_{x \rightarrow 0} f(x, 0) = 0 = \lim_{y \rightarrow 0} f(0, y)$

is this enough to say limit exists and is equal to 0?

Check along the line  $y=mx$

$$\lim_{x \rightarrow 0} f(x, mx) = \lim_{x \rightarrow 0} \frac{x \cdot (mx)^2}{x^2 + m^4 x^4} = \lim_{x \rightarrow 0} \frac{m^2 x}{1 + m^4 x^3} = 0$$

Now this is enough to say limit exists and equal 0.



try with another curve or path to approach point  $(0,0)$ ! So +  
consider parabola  $x = my^2$

$$\lim_{y \rightarrow 0} f(my^2, y) = \lim_{y \rightarrow 0} \frac{my^4}{m^2y^4 + y^4} = \frac{m}{m^2 + 1}$$

which is diff. for different value of  $m$ .

$$\text{so } \lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{x^2 + y^4} \text{ D.H.G.}$$

Using the  $\epsilon, \delta$  definition to prove existence of a limit

Ex Prove that  $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0$  where  $f(x,y) = \frac{3xy^2}{x^2 + y^2}$

Why should we expect that this limit exists?

Hint: The numerator is cubic and denominator is quadratic  
so we can guess who should win in a long run.

Proof

$$\begin{aligned} |f(x,y) - 0| < \epsilon &\Rightarrow \left| \frac{3xy^2}{x^2 + y^2} - 0 \right| < \epsilon \\ &\Rightarrow \left| \frac{3xy^2}{x^2 + y^2} \right| < \epsilon \\ &\Rightarrow 3|x| \frac{y^2}{x^2 + y^2} < \epsilon \end{aligned}$$

We know

$$x^2 \leq x^2 + y^2 \text{ and } 0 \leq \frac{y^2}{x^2 + y^2} \leq 1$$

We have

$$3|x| \frac{y^2}{x^2 + y^2} \leq \underline{3|x| \frac{y^2}{x^2 + y^2}}$$

$$\Rightarrow 3\sqrt{x^2} \leq 3\sqrt{x^2 + y^2}$$

$$\text{Now choose } \delta = \frac{\epsilon}{3}$$

So we have whenever  $\sqrt{x^2+y^2} < \delta = \frac{\epsilon}{3}$   
the inequality  $3\sqrt{x^2+y^2} < \epsilon$  holds.

Meaning

$$|f(x,y) - 0| \leq 3|x| \frac{y^2}{x^2+y^2} \leq 3\sqrt{x^2+y^2} < \epsilon$$

this proves that

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0$$

### Repeated limits of $f(x,y)$ at $(a,b)$

$$\lim_{x \rightarrow a} \lim_{y \rightarrow b} f(x,y)$$

and

$$\lim_{y \rightarrow b} \lim_{x \rightarrow a} f(x,y)$$

Ex consider func<sup>n</sup>  $f(x,y) = \frac{x^2}{x^2+y^2}$

$$\begin{aligned} \lim_{y \rightarrow 0} \left( \lim_{x \rightarrow 0} \frac{x^2}{y^2+x^2} \right) &= \lim_{y \rightarrow 0} \left( \frac{0}{y^2+0} \right) \\ &= \lim_{y \rightarrow 0} \left( \frac{0}{y^2+0} \right) = \lim_{y \rightarrow 0} (0) = 0 \end{aligned}$$

$$\lim_{x \rightarrow 0} \left( \lim_{y \rightarrow 0} \frac{x^2}{y^2+x^2} \right) = \lim_{x \rightarrow 0} \left( \frac{x^2}{x^2} \right) = 1$$

Ex  $f(x,y) = \frac{xy}{x^2+y^2}$

$$\lim_{y \rightarrow 0} \left( \lim_{x \rightarrow 0} \frac{xy}{x^2+y^2} \right) = \lim_{y \rightarrow 0} (0) = 0$$

$$\lim_{x \rightarrow 0} \left( \lim_{y \rightarrow 0} \frac{xy}{x^2+y^2} \right) = \lim_{x \rightarrow 0} (0) = 0$$

} Now they are equal.  
But about simultaneous limit?

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = \lim_{(x,y) \rightarrow (0,0)} \left( \frac{xy}{x^2+y^2} \right) \quad \text{along } y=mx$$

$$= \lim_{(x,y) \rightarrow (0,0)} \left[ \frac{x \cdot mx}{x^2 + x^2 m^2} \right]$$

$$= \frac{m}{1+m^2} \quad \text{depending on } m \text{ so DNE.}$$

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## Continuity of func<sup>n</sup> of Several Variables

Let  $f$  be a real valued function defined in a ball around  $(a,b)$ . Then  $f$  is said to be continuous at  $(a,b)$  if

$$\lim_{(x,y) \rightarrow (a,b)} f(x,y) = f(a,b)$$

Q

$$f(x,y) = \begin{cases} \frac{xy}{\sqrt{x^2+y^2}} & x^2+y^2 \neq 0 \\ 0 & x=y=0 \end{cases}$$

check continuity at  $(0,0)$

$$\text{Let } \epsilon > 0. \text{ Then } |f(x,y) - 0| \leq \left| \frac{xy}{\sqrt{x^2+y^2}} \right| = |x| \frac{|y|}{\sqrt{x^2+y^2}} \leq |x|$$

$$\text{choose } \delta = \epsilon \text{ then } |f(x,y) - 0| \leq \epsilon \quad \text{if } |x| < \delta$$

so c<sup>t</sup> at  $(0,0)$

$$Q \quad f(x,y) = \begin{cases} x^2 + 2y & (x,y) \neq (1,2) \\ 3 & (x,y) = (1,2) \end{cases}$$

$f$  is not continuous at  $(1,2)$  since

$$\lim_{(x,y) \rightarrow (1,2)} x^2 + 2y = 5 \neq f(1,2)$$



## Partial Derivatives

Let  $f = f(x, y)$  then partial derivative of  $f$  with respect to  $x$  at  $(a, b)$  is

$$f_x(a, b) = \frac{\partial f}{\partial x}(a, b) = \lim_{h \rightarrow 0} \frac{1}{h} [f(a+h, b) - f(a, b)]$$

wrt to  $y$  is

$$f_y(a, b) = \frac{\partial f}{\partial y}(a, b) = \lim_{k \rightarrow 0} \frac{1}{k} [f(a, b+k) - f(a, b)]$$

Q

$$f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

not continuous at  $(0, 0)$  but partial derivative exists.

Ans

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0 - 0}{h} = 0$$

$$f_y(0, 0) = \lim_{k \rightarrow 0} \frac{f(0, k) - f(0, 0)}{k} = \lim_{k \rightarrow 0} \frac{0 - 0}{k} = 0$$

$$\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = \lim_{(x, y) \rightarrow (0, 0)} \frac{xy}{x^2 + y^2} \quad \text{along } y = mx$$

$$= \lim_{(x, y) \rightarrow (0, 0)} \frac{x(mx)}{x^2 + m^2x^2} = \frac{m}{1+m^2}$$

$$f(0, 0) = 0$$

$$\lim_{(x, y) \rightarrow (0, 0)} f(x, y) \neq f(0, 0) \quad \forall m \Rightarrow \text{not continuous}$$

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②  $f(x,y) = |x| + |y|$  ~~is not a function~~ ~~partial derivative~~  
DNE.

$$f_x(0,0) = \lim_{h \rightarrow 0} \frac{f(0+h,0) - f(0,0)}{h} = \lim_{h \rightarrow 0} \frac{|h|}{h} = \text{DNE.}$$

$$f_y(0,0) = \lim_{k \rightarrow 0} \frac{f(0,0+k) - f(0,0)}{k} = \lim_{k \rightarrow 0} \frac{|k|}{k} = \text{DNE.}$$